Hive, HiveQL, Hbase and Zookeeper

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What is Hive?:

Hive is a data warehousing system built on top of Hadoop, designed for querying and analyzing large datasets. Its architecture consists of several key components:

- **User Interface (UI)**
- **Driver**
- **Compiler**
- **Metastore**
- **Execution Engine**
- **HDFS (Hadoop Distributed File System)**

HiveQL - Querying Data: Sorting and Aggregating:

HiveQL provides SQL-like syntax for querying data.

- Sorting Data:
  ORDER BY: is a SQL clause used to sort the results of a query by one or more columns in either ascending (ASC) or descending (DESC) order.
  SORT BY: Sorts data within each reducer, providing a partial order. This is more scalable for large datasets as multiple reducers can process data in parallel.
- Aggregating:
  HiveQL supports various aggregate functions for summarizing data.
  COUNT() -> Counts the number of rows or non-null values.
  SUM() -> Calculates the sum of a numeric column.
  AVG() -> Calculates the average of a numeric column.
  MIN()/MAX() -> Finds the minimum or maximum value in a column.

HBase Concepts:

HBase is a distributed, column-oriented NoSQL database built on top of the Hadoop Distributed File System (HDFS). It is designed to provide real-time, random read/write access to massive datasets, making it a key component in Big Data ecosystems.

Apache HBase is an open-source, distributed, versioned, non-relational database modeled after Google's Bigtable: A Distributed Storage System for Structured Data.

Features : Horizontally scalable across many nodes and Strictly consistent for reads and writes.

Automatic and configurable sharding of tables and Automatic failover support between RegionServers.

Convenient base classes for backing Hadoop MapReduce jobs with Apache HBase tables.

Easy to use Java API for client access.

Extensible jruby-based (JIRB) shell.

HBase Architecture:

- HMaster
- RegionServers
- Region
- ZooKeeper

Advance useage of HBase :

- HBase supports real-time analytics, data streaming, time-series applications, integration with other Big Data tools, replication, snapshots.

Indexing in HBase :

In HBase, data is automatically indexed by its Row Key, meaning every row in a table is stored in sorted order based on the Row Key.

Types of Indexes in HBase

- Primary Index
- Secondary Index
- Composite Index

Differences Between Hive and HBase

S.No	Feature / Aspect	Hive	HBase
1	Type / Nature	Data warehouse on Hadoop	NoSQL, column-oriented database
2	Purpose	Batch processing and data analytics	Real-time read/write access
3	Processing Type	Batch (MapReduce/Spark)	Real-time (RegionServers)
4	Data Model	Table-based (like RDBMS)	Column-family based (NoSQL model)
5	Query Language	HiveQL (SQL-like)	API (Java) or Phoenix SQL

Differences Between Hive and HBase

S.No	Feature / Aspect	Hive	HBase
6	Schema Type	Schema-on-Read (flexible)	Schema-on-Write (predefined)
7	Data Access Pattern	Sequential scan (analytical)	Random access (record-level)
8	Speed / Latency	Slower (batch)	Faster (real-time)
9	Use Case	Reporting, summaries, analytics	Online transactions, IoT, user data
10	Integration	Can query HBase data via HBaseStorageHandler	Acts as backend for Hive tables